

Integrations as the potential targets for combating multidrug resistance in *Enterobacteriaceae* using CRISPR- Cas9 technique

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Abstract

The emergence of multi-drug resistance (MDR) to pan-drug resistance (PDR) in *Enterobacteriaceae* has made treatment extremely challenging. Genetic mutations and horizontal gene transfer (HGT) through mobile genetic elements (MGEs) were frequently associated mechanisms of drug resistance in pathogens. However, transposons, plasmids, and integrations transfer MDR genes in bacterium via HGT much faster. Integrations are dsDNA segment that plays a crucial role in the adaptation and evolution of bacteria. They contain multiple gene cassettes that code for antibiotic resistance determinants that are expressed by a single promoter (Pc). Integrations are the cause of drug resistance in *Enterobacteriaceae*. Although alternatives to antibiotics such as bacteriophages, phage proteins, antimicrobial peptides, and natural compounds have been widely used to treat MDR infections, there have been limited efforts to reverse the antibiotic resistance ability of bacteria. Thus, silencing the genes harboured on MGEs achieved by Gene Editing Techniques (GETs) might prevent the spread of MDR. One such GETs, which has a simple design, good repeatability, low cost, and high efficiency, is CRISPR- Cas9 system. Thus, this review is a first of the kind that focuses on utilizing the structure of an integration to make it an ideal target for GETs like CRISPR- Cas9 systems.

Keywords: integrations, multidrug resistance, CRISPR- Cas, mobile genetic elements, gene editing techniques

Introduction

Enterobacteriaceae is a major family that causes varied human diseases within the large group of Gram-negative bacteria. It includes *Salmonella* spp., *Klebsiella* spp., *Citrobacter* spp., *Enterobacter* spp., *Escherichia coli*, *Serratia* spp. and other species (Janda et al. 2021; Tamma et al. 2022). It has become a challenge to treat common infections with antibiotics due to the frightening spread of resistance. Also, it causes a variety of severe conditions, including enteric diseases, cystitis, acute pyelonephritis, community-acquired pneumonia, ventilator-associated pneumonia, etc., which can sometimes be severe, and life-threatening if not treated properly (Pitout 2010; Sheu et al. 2019; Tamma et al. 2022). The increase in morbidity and mortality in *Enterobacteriaceae* infections is mainly associated with the diseases caused by MDR, Extensively drug-resistant (XDR), Pan drug resistance (PDR) strains due to inappropriate treatment and limited choice of drug to treat the infections (Delgado et al. 2013; Potter et al. 2016). Further, diseases caused by *Enterobacteriaceae* from the hospital environment are becoming less susceptible to effective antibiotics limiting the treatment options. Current reports of carbapenem-resistant *Enterobacteriaceae* (CRE) show a considerable impact on the health sector since the strains show resistance to nearly all the available antibiotics leading to the development of MDR and PDR strains (WHO 2017; Abe et al. 2021). A recent ground-breaking report by United Nations increased the uncertainties on antibiotic resistance. It predicted that deaths could rise to 10 million by the year 2050, forcing the developed and under-developed countries'

populations into extreme poverty (UN 2019). CRE global dissemination due to geographical expansion and the spread of *bla*_{VIM}, *bla*_{KPC}, *bla*_{CTX}, *bla*_{VIM}, *bla*_{IMP}, *bla*_{NDM}, *bla*_{Cmy2}, *bla*_{Dha1}, *bla*_{Fox1}, and *bla*_{OXA-48} gene variants has worsened the possibilities for treating the *Enterobacteriaceae* infections (Delgado et al. 2013; Tzouveleakis et al. 2014). The rate of development of resistance is far more than the formulation of new antibiotics. There are several mechanisms of resistance, for instance, modifications of the antibiotic molecule, efflux, alteration in target sites, resistance due to global cell adaptations, and decreased antibiotic permeability through outer membrane proteins (Deekshit et al. 2015; Akshay et al. 2022).

Antibiotic resistance can either be mediated by chromosomal or plasmid-borne genes; however, the mobile genetic elements (MGEs), such as plasmids, transposons and integrations play a critical role in the transmission of resistance by capture, accumulation, and dissemination of resistance in bacteria (Domínguez-Gil et al. 2016; Partridge et al. 2018; Akshay et al. 2023).

MGEs contain unique backbone genes that enable them to replicate the cellular chromosome independently via homologous or non-homologous recombination. They also extrude plasmids or package bacteriophage DNA for effective movement between cells (Frost et al. 2005; Feng et al. 2023). Integrations are one such MGEs that are directly associated with spreading antibiotic resistance through plasmids; It is a primeval part of bacteria that helps in evolution by storing, resorting, and disposing of genes to the reading frameworks in gene cassettes (Sabbagh et al. 2021). They have a crucial role

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in horizontal gene transfer (HGT) and promote the spread and acquisition of antibiotic resistance genes (ARGs) either within a conjugative plasmid or in chromosomes (Kaushik et al. 2018). Integrons are flexible and act as scaffolds for rearranging multiple genes by site-specific recombination of gene cassettes. Due to its ability to accommodate many ARGs, it is threatful for spreading drug resistance in nosocomial pathogens.

Currently, the approaches to treat MDR or PDR infection can efficiently cope with genome editing technologies (GETs) (Li et al. 2020). There are several GETs, including transcription activator-like effector nucleases (TALENs), Zinc-finger nucleases (ZFNs), and RNA-guided CRISPR-Cas nuclease system (Clustered Regularly Interspaced Palindromic Repeats-CRISPR) that triggers a significant turmoil in biomedical research. The first two GETs confine endonuclease catalytic domains to modular DNA binding proteins for inducing targeted DNA double-stranded breaks (DSBs) at specific genomic loci. However, RNA-guided CRISPR-Cas is a nuclease directed by small RNAs through Watson and Crick base pairing with the target. Thus, CRISPR-Cas systems are more beneficial due to more comfortable to design, well-suited for high throughput, and highly precise and multiplexed gene editing for various cell types and organisms (Ran et al. 2013, Redman et al. 2016).

Hence, this review aims to augment the advanced knowledge of different classes, structures, and functions of integrons and is a humble attempt to recapitulate the potential of CRISPR-Cas to repress the antibiotic-resistant integrons that disseminate resistance in *Enterobacteriaceae*.

Structure and classes of integrons

General structure of integron

Integrons are genetic elements that rapidly acquire and rearrange mobile genes, called gene cassettes (Ghaly et al. 2022). Gene cassettes captured by site-specific recombination are a significant component of integrons. These cassettes carry genes that are acquired, encoded, or deleted and thus contribute to the varying characteristics of the bacteria (Sabbagh et al. 2021, Barraud et al. 2022).

Generally, the major components of integrons are *intI*, *attI* (integron attachment site), and *Pc/PintI* (Akrami et al. 2019). A study reported that the *intI* gene codes for an integrase protein *intI* that belongs to the tyrosine kinase family (Faghri et al. 2018). Also, the C-terminal catalytic domain of *intI* contains Arg146-Lys171-His277-Arg280-His/Trp303 conserved amino acids and the nucleophilic tyrosine (Tyr312), which is essential for the site-specific recombination between *attI* and *attC* (gene cassette recombination site) (Ghaly et al. 2021). Cassettes are inserted or excised due to *IntI*-catalyzed recombination between *attI* and/or *attC* sites (Partridge et al. 2018). *Pc/P_{intI}* is the promoter region that enables the expression of the genes in cassettes (Ghaly et al. 2022). The position of the gene cassettes with respect to the promoter of the integron is responsible for determining the level of expression of the genes in the cassette. Close proximity of the genes towards the promoter usually increases gene expression (Lu et al. 2022).

59-base elements

59-base elements are cassette-associated recombination sites located downstream of the gene cassette. Initially recognized

as imperfect inverted repeat sequences, these elements vary in length (57–141 bp) and hence are not highly conserved. They are identified by their position and relationship of over 20 bp at their outer ends to consensus sequences (Singh et al. 2022). They are present on either end of each gene cassette (Najafi et al. 2022). A variable region exists between the 20 bp consensus sequence; hence they carry a 2-fold axis of symmetry. One of the critical features of these elements is that either end contains highly conserved sequences; GTTAGGC or GTTRRRY at the right-hand (RH) end of the element and GCCTAAC or RYYYAAC at the left-hand (LH) end of the element (Stokes et al. 1997). The cassette insertion in the correct orientation is determined by the element's difference in RH and LH sequence.

Gene cassettes

Besides the three key integron components, gene cassettes play a major role in the transmission of antibiotic resistance. Gene cassettes are the minimal functional components designed to be mobilised by the integrase of integrons. They consist of a simple structure; a 59-base element recognized by the integron-encoded site-specific recombinase (*IntI*) and an open reading frame (ORF) (Tansirichaiya et al. 2019). These mobile elements exist in either free or circular form, circular when not inserted in a cassette array (Mazel 2006, Gillings 2014). The gene cassettes are randomly assembled between integrons conserved 3' and 5' segments. These cassettes become a part of integron when they integrate themselves at integron receptor site (Richard et al. 2022). The integrase gene uses two recombination sites (*attI*, *attC*) to integrate gene cassettes. This process is reversible, and cassettes are released from integrons as free DNA (Xu et al. 2011). A single site-specific recombination event, including the integron-associated *attI* site and a cassette-associated 59-base element, leads to inserting a free circular gene cassette into a recipient integron (Singh et al. 2022). All the gene cassettes in an integron will be transcribed through a common single promoter (Pereira et al. 2020). The details of gene cassettes coding for different antibiotic resistance genes are being summarized in Table 1.

Classification

Earlier, based on their mobility, it was proposed that integrons could be classified into two major categories, mobile integrons, and super integrons (Wu et al. 2013). Mobile integrons consist of ARGs with different *attC* sites, with a few gene cassettes and depend on plasmids or transposons for mobility. Super integrons usually found on the chromosomes have similar *attC* sites and multiple gene cassettes with multiple promoters (Ravi et al. 2014). Ever since the study on integrons has speeded up, they can now be classified into different classes based on the amino acid sequences of the integrase gene (Kaushik et al. 2018). Currently, integrons can be classified into five different classes (Class 1 integron containing *intI1* integrase gene, Class 2 integron containing *intI2* integrase gene, Class 3 integron containing *intI3* integrase gene and so on (Sabbagh et al. 2021, Néron et al. 2022). However, class 1, 2, and 3 integrons were initially identified to be associated with mobile genetic elements, while class 4 and 5 were chromosomal integrons (Deng et al. 2015, Partridge et al. 2018). Amongst these classes, class 1 was found to be more prominent in clinical isolates and is the major player in the

Table 1. Depicts different gene cassettes associated with different classes of integrons in *Enterobacteriaceae*

Different <i>Enterobacteriaceae</i>	Integron class	Gene cassettes associated with it	Antibiotics associated with the gene cassette	Protein Encoded	Reference
<i>E. coli</i>	Class 1	<i>aadA1, aadA2, aadA5, aacA4, catB3, dfrA1, dfrA17, dhfr12, dfrA12, dfrA7, bla_{OXA-1}, bla_{CARB-2}, ereA</i>	Aminoglycosides, Phenicol, Dihydrofolate reductase, β -lactams, Macrolides	Adenyltransferase, Chloramphenicol acetyltransferase, Dihydrofolate reductase, Extended spectrum β lactamase, Erythromycin esterase	Zhang et al. 2019, Blancarte Lagunas et al. 2020, Huang et al. 2020, Mohamed et al. 2020, Kalantari et al. 2021, Singh et al. 2021, Behera et al. 2023
	Class 2	<i>aadA1, dfrA1, sat2</i>	Aminoglycosides, Dihydrofolate reductase	Adenyltransferase, Dihydrofolate reductase, streptothricin acetyltransferase	Zhang et al. 2019, Singh et al. 2021
	Class 3	<i>tetA, aadA1, aadA2, bla_{OXA-10}, bla_{GES-1}</i>	Tetracycline Aminoglycosides, β -lactams, Anti-metabolites	Tetracycline resistance protein, Adenyltransferase, Extended spectrum β lactamase	Mohammadaliipour et al. 2017
<i>Salmonella</i>	Class 1	<i>aadA1, aadA2, aadA5, dfrA1, dfrA17, dfrA12, bla_{OXA-30}, bla_{PSE}, bla_{PSE-1}, bla_{CTXM-15}, bla_{TEM-1}, qacEΔ1, sul1, tet, aadB, catB3, aadA22</i>	β -lactams, Quaternary ammonium compounds, Sulphonamide, Tetracycline, Aminoglycosides, Dihydrofolate reductase, β -lactams, Phenicol, Tetracycline, Phenicol, Lincosamides, Aminoglycosides	Adenyltransferase, Dihydrofolate reductase, Extended spectrum β lactamase, Chloramphenicol acetyltransferase, quaternary ammonium compound efflux SMR transporter QacE delta 1, sulfate permease, tetracycline resistance protein	Ahmed et al. 2005, Amador et al. 2019, Zhang et al. 2019, Rao et al. 2020, Ren et al. 2020, Sin et al. 2020, Zhao et al. 2020, Li et al. 2021
	Class 2	<i>tet, catA1, linE, estX</i>	β -lactams, Phenicol, Tetracycline, Phenicol, Lincosamides, Aminoglycosides	Chloramphenicol acetyltransferase, Streptothricin acetyltransferase, tetracycline resistance protein, lincosamide nucleotidyltransferase	Ahmed et al. 2005, Zhang et al. 2019, Ren et al. 2020; Argüello et al. 2018.
<i>Klebsiella</i>	Class 1	<i>aacA4, bla_{OXA2}, aadA2, aadA1, dfrA1, bla_{GES-1}, aac, dfrXVb, cmlA4, dfrA12, dfrA5, dfrA30</i>	β -lactams, β -lactamase inhibitor, Aminoglycosides, Phenicol, Dihydrofolate reductase	Adenyltransferase, Dihydrofolate reductase, Extended spectrum β lactamase, chloramphenicol efflux transporter	Poirer et al. 2000, Rao et al. 2006, Roy Chowdhury et al. 2011, Li et al. 2013, Lima et al. 2014
	Class 2	<i>aadA1, dfrA1-sat1</i>	Aminoglycosides, Dihydrofolate reductase	Adenyltransferase, Dihydrofolate reductase, streptothricin acetyltransferase	Poirer et al. 2000, Lima et al. 2014
	Class 3	<i>bla_{OXA-10}, bla_{GES-1}</i>	Dihydrofolate reductase β -lactams	Extended spectrum β lactamase	Correia et al. 2003, Firoozeh et al. 2019
<i>Shigella</i>	Class 1	<i>dfrA17, aadA5, dfrA12, aadA2, arr-3, aacA4, sul1, sul2</i>	Dihydrofolate reductase, Aminoglycosides, sulphonamide	Dihydrofolate reductase, Adenyltransferase, sulfate permease	Peirano et al. 2005, Jing Yuan et al. 2011, Yang et al. 2014, Shahin et al. 2019
	Class 2	<i>dfrA1, sat1, aadA1</i>	Aminoglycosides, Dihydrofolate reductase	Adenyltransferase, Dihydrofolate reductase, streptothricin acetyltransferase	Jing Yuan et al. 2011, Yang et al. 2014, Shahin et al. 2019
<i>Citrobacter</i>	Class 1	<i>dfrA14, dhfrA1, arr2, cmlA5, bla_{OXA-10}, bla_{OXA-1}, aadA, aac(6=)-Ib-cr, catB3, arr3, aadA1, bla_{NDM-1}, sul1, sul2, mph(A), aadA16, dfrA16, mph, vim-1</i>	Dihydrofolate reductase, aminoglycosides, macrolide, Phenicol, sulphonamide, β -lactams	Adenyltransferase, Dihydrofolate reductase, chloramphenicol efflux transporter, chloramphenicol acetyltransferase, sulfate permease, macrolide phosphotransferase A, metallo β lactamase, fluoroquinolone-acetylating aminoglycoside 6'-N-acetyltransferase	Lee et al. 2016, Lorestani et al. 2018, Hasan et al. 2019, Li et al. 2020
	Class 2	<i>dfrA1-sat2-aadA1</i>	Aminoglycosides, Dihydrofolate reductase,	Adenyltransferase, dihydrofolate reductase, streptothricin acetyltransferase	Xia et al. 2013.

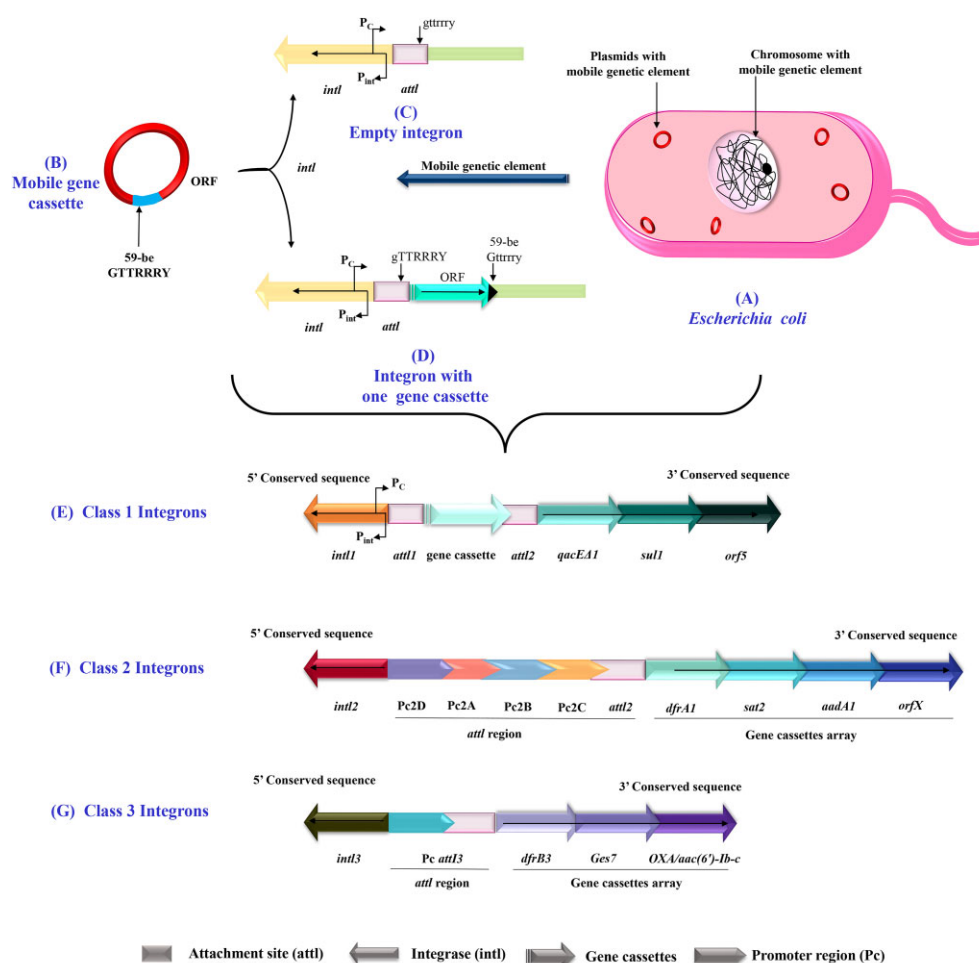


Figure 1. Schematic representation of *E. coli* integrons and their three different classes with gene cassettes. (A) *E. coli* containing the integrons within the plasmid and as an integral part of chromosome (B) Mobile genetic element with 59-bp recombination site and ORF (C) An empty integron composed of specific integron integrase gene (*intI*) that encodes the integrase, promoters *P_{int}* and *P_C*, and an attachment recombination site (*attI*) (D) Complete integrons containing one gene cassette with 59-bp recombination site. (E) Class 1 integron (F) Class 2 integron (G) Class 3 integron.

spread of antibiotic resistance; hence they are being widely investigated (Sabbagh et al. 2021).

Class 1 integron

Corynebacterium glutamicum was the first organism to contain a class I integron, which was later discovered in *Streptococcus*, *Enterococcus*, *Aerococcus*, *Staphylococcus*, and *Brevibacterium* (Deng et al. 2015).

About 9% of the bacterial genomes sequenced consist of integrons, and the class 1 integron platform is the most prevalent one (Barlow et al. 2004). Based on the amino acid homology of the integron-integrase protein, the distinctive family of class 1 integrons are related. As a common contributor to the vast dispersal and dissemination of antimicrobial resistance, class 1 integrons are frequently used as substitutes for anthropogenic pollution (Sanchez et al. 2023). In the recent times, they possess great clinical importance. It has been the most frequently reported integron among clinical bacteria and is still the subject of many investigations (Filipić et al. 2023). Most enteric bacteria, including all common disease-causing human pathogens like *E. coli*, *Salmonella*, *Shigella*, *Klebsiella*, and other pathogenic bacteria, are known to harbour class 1 integrons (Kaushik et al. 2019; Abbasi et al. 2023).

A typical class 1 integron consists of ARGs within the gene cassettes between two conserved regions, namely 5' CS (conserved sequences) and 3' CS. The 5' CS consists of *intI* gene, promoter *P_C* and *attI* site with *P_C* about 200 bp upstream of the integration site (Deng et al. 2015; Fonseca et al. 2022). The 3' CS includes *sul1* gene encoding sulphonamide resistance; *qacEΔ1* gene for quaternary ammonium compound resistance; and an ORF *orf5*, whose function remains unknown (Chaturvedi et al. 2021). Gene cassettes are incorporated accurately between G and TT in the right hand core site (Gttrrry) of the *attI* simple site (Najafi et al. 2022) (Fig. 1). There is a considerable variation in the sequence of the gene cassettes of class 1 integron within the *attC* that results in the insertion of any gene cassette belonging to the distinct genomic framework (Ghaly et al. 2021). Remarkably, the independent and the transposable nature of new gene cassettes attached can lead to tremendous genetic diversity while preserving the integrity of the genome. The reaction between *attI* and 59-bp requires an area of 65 bp consisting of a simple site. It comprises two inversely oriented *IntI* binding domains (Fig. 1). Class 1 integrons are not self-movable and are considered to be closely related to *Tn402*-like transposons and members of the *Tn3* transposon family (*Tn21* or *Tn1696*) within a plasmid (Deng et al. 2015). This class of integron has been found to contain

>130 different gene cassettes that give resistance to most of the known cephalosporins, carbapenems, monobactams, penicillins, aminoglycosides, trimethoprim, acyclovir, chloramphenicol, fluoroquinolones, and quaternary ammonium compounds (Deng et al. 2015, Malek. 2015; Li et al. 2023). The majority of the class 1 integrans carry few antibiotic resistant genes in common that includes sulphonamide resistant genes (*sul*), trimethoprim resistant genes (*dfrA*), chloramphenicol resistant genes (*cat*), aminoglycoside resistant genes (*aadA*), rifampin resistant genes (*rpo*), quaternary ammonium compounds (*qacE Δ1*) (Akrami et al. 2019). Additionally, class 1 integrin diversification and spread are aided by human activity. *sul1*, which is significantly related with clinical integrans, is more prevalent in sewage in proportion to the number of antibiotic prescribed, showing that antibiotic selection directly influences the prevalence of *sul1* in human populations (Gillings 2017). While talking of the prevalence of these integrans among different sources, Gillings (2018), studied that this class of integrin was highly found in the chicken manure and pharmaceutical plant sediments; and was least present in the soil. The extensive distribution of *intI1* in these areas clearly implies that *intI1* is the subject of active selection and gives the bacteria that carry it a survival advantage. Especially, the passage of the waste water into rivers from sewage treatment plant is a major route of dissemination of *intI1*. However, no significant difference was observed in the mean value of the *intI1* presence within sewage water effluent, solid waste of treatment plant, waste water and river water. This directly indicates that treatment of waste water do not decrease the magnitude of bacteria harbouring class 1 integrin (Gillings 2018). Though the class 1 integrans have allowed a crucial understanding of the structure and function of genetic elements within its construct and were widely associated with most drug-resistant bacteria, they do not represent integrans as a whole. When considering the evolution and structural details of integrans, it is essential to consider the different classes of integrans, looking further for more diversity of genes.

Class 2 integrin

Class 2 integrans have a unique integrase gene (*intI2*) with an internal stop codon at the 179th position, leading to a non-functional protein (Jones et al. 2016). Due to the absence of integration activity of the *intI2* gene, which is frequently truncated, class 2 integrans typically exhibit less variability than class 1 and class 3 (Jové et al. 2017). Due to this feature, it can accommodate very few antibiotic-resistance gene cassettes such as *sat2* encoding resistance to streptothricin, *dfrA1* conferring resistance to trimethoprim and *aadA1* encoding resistance for streptomycin (Najafi et al. 2022). Nevertheless, there have been two resistant gene determinants identified recently: ($\Delta attC$)-*dfrA1-sat2-aadA1-orfX* and *sat2-aadB-catB2* (Shariati et al. 2018). Despite the difference in integrase genes, they share a structure with class 1 integrans in the 5' regions (TAA). A 46% similarity exists between the integrase genes in class I and class II integrans (Sabbagh et al. 2021). Eventhough, class 2 integrans have the ability to excise and integrate gene cassettes into *attI2* site, *intI2* fails to recognize *attC* sites from class 1 integrans (Deng et al. 2015). There are relatively few class 2 integrans characterised with a non-disrupted *intI2* gene expressing a 325-aa full-length functional integrase. In addition, replacing codon TAA at the 179th amino acid posi-

tion with glutamic acid yields an active integrase (Barlow and Gobius 2006, Wang et al. 2021). They are associated with the *Tn7* transposon family for their mobility (Benler et al. 2021). The presence of five *tns* genes (*tnsA*, *tnsB*, *tnsC*, *tnsD*, and *tnsE*) in its 3' conserved segment (3'-CS) facilitates the transposon movement through preferential insertion into a specific location on bacterial chromosomes (Labate et al. 2009, Choi et al. 2014). With the unique feature of an internal stop codon, class 2 integrans exhibit diversity amongst the rest of the classes of integrans though they are structurally similar to class 1 integrans. However, the prevalence of class 2 integrans is much higher than class 3.

Class 3 integrin

In 1993, Japanese scientists discovered Class 3 integrin from isolates of *Serratia marcescens* (Deng et al. 2015). Since then, it has been identified in very few microorganisms, including *E. coli*, *Pseudomonas putida*, *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, *S. marcescens*, *Citrobacter freundii* and *Salmonella* spp (Mohammadaliipour et al. 2017; Elmonsef et al. 2022). The class 3 integrans resemble class 1 and 2 in terms of their structure. It resembles to class 1 integrase gene by 60% (Kaushik et al. 2018). It has been discovered that *intI3* can catalyse the integration of circularised cassettes containing different *attC* sites with lower recombination frequencies than those seen with *IntI1* (Deng et al. 2015). They can also catalyse integrated cassette removal from the *attI3* site (Loot et al. 2022). The common genes that were found to be associated with class 3 integrans are IMP-1 metallo- β -lactamase and *bla*_{GES-1} within the IncQ plasmid from *E. coli* (Akrami et al. 2019).

Class 4 integrans

As mentioned earlier, class 4 integrans are usually associated with chromosomal regions of microorganisms. They were first identified in the year 2002 on a small chromosome of *Vibrio cholera* consisting of 179 gene cassettes and found to be an integral part of many γ -proteobacterial genomes (Li et al. 2018). It is distinct from other classes of integrans by the number of gene cassettes incorporated and the high homology between *attC* sites of gene cassettes. It has been identified to date only in a few groups of microorganisms such as *Vibrio*, *Shewanella*, *Xanthomonas*, *Pseudomonas*, and other proteobacteria (Poirel et al. 2010, Sabbagh et al. 2021). However, not much studies have been carried out on class 4 integrans.

Class 5 integrans

At the moment, due to the complex nature of integrans and their highly conserved class-specific genes and cassettes, one such class of integrans that have been classified as per the genetic diversity is said to be class 5 integrans; also previously termed as chromosomal super integrans (Hochhut et al. 2001, Kaushik et al. 2019). Due to less abundance in the environment, this integrin class has not been studied extensively, and there are only a few reports on the occurrence of class 5 integrans in *Enterobacteriaceae*. However, it has been detected on a plasmid pRVS1 of a *Vibrio salmonicida* strain. It is usually found to be associated with only one antibiotic-resistance gene cassette and several non-resistance gene cassettes (Rowe Magnus and Mazel 2002).

Evidences of integrons

Even though the existence of integrons is not new, their closer association with antibiotic resistance has made researchers look back at them. Divided into different classes, each integron class has its diversity from existence to dissemination. Many studies depict the presence of class 1 integrons from various sources, including environmental and clinical. The high prevalence of class 1 integrons in the environment facilitated them to invade other human pathogens of medical importance (Gillings 2017, Zhang et al. 2018). This further allowed them to spread to different anthropogenic environments such as hospital wastewater (Guo et al. 2011, Stalder et al. 2014, Marathe et al. 2019), municipal wastewater, animal wastes, compost, manure soils and biotreatment systems (Gaze et al. 2011, Koczura et al. 2012, Stalder et al. 2014, Ma et al. 2017, Wang et al. 2020).

Their occurrence in food materials has raised the concern of them being contagious to humans. In agreement with this, several studies have indicated the presence of different classes of integron-spreading antibiotic-resistance genes. A study on ready-to-eat vegetables in Indonesia revealed the presence of class 1 integron in MDR *E. coli* isolates, wherein 38% of these bacterial populations contained *dfrA7* gene cassette that conferred resistance to trimethoprim and about 11% of the isolates had integrons without any gene cassette, marking themselves as potential carriers of antibiotic-resistance genes (Mohamed et al. 2020). Even though ready-to-eat foods are most preferred, integrons contributing to MDR within the bacteria are frightening (Zhang et al. 2020). Likewise, the occurrence of class 1 integrons in *Vibrio parahaemolyticus* was encountered in a study on ready-to-eat foods in China. Out of 51 samples tested, all were positive for integrons, and two MDR isolates contained *bla_{VEB1}* gene (Lei et al. 2017). Correspondingly, A study conducted by Jenifer and Sathiyamurthy (2020) on ready-to-eat food products procured from Tamil Nadu, India had screened for class 1 integrons in *E. coli*, where they found that 69.1% of isolates were MDR and 58.3% of them were identified with the *intI1* gene. However, integrons are also notably present in water sources along with ready-to-eat foods. Water aids several purposes in our daily life, making aquatic organisms a possible conveyor of drug-resistant integrons. In accordance with this statement, a study conducted on urban river water in India, denoted the presence of class 1 integrons in 32% of the MDR *E. coli* isolates that harboured several resistance genes, including *dfrA17-aadA5*, *dfrA12-orfF-aadA2*, *bla_{OXA-1}-aadA1* (Kaushik et al. 2019). Similarly, Canal et al. (2016) reported the genetic diversity in *E. coli* isolated from the surface water of Patos Lagoon, Brazil, resistance to trimethoprim, quaternary ammonium compounds, streptomycin/spectinomycin conferred by *dfrA17*, *dfrA1*, *dfrA12*, *qacEΔ1*, *aadA1*, *aadA5*, and *aadA22*. Also, data from China highlighted the health threat posed by ARGs in the drinking water supply when investigated in well and tap water from four cities in Henan Province. In both well and tap water, ARG and *intI1* expression exhibited strong correlations (Zhang et al. 2021). Nonetheless, the detection of class 1 integrons conferring resistance to co-trimoxazole in seafood isolates of *Salmonella* Weltevreden and *Salmonella* Newport has also been emphasized (Khan et al. 2009, Deekshit et al. 2012). A similar report highlighted that 30% *P. aeruginosa* isolates harboured class 1 integrons that showed 100% resistance towards cephalosporins and cephalosporins and 13.3% resis-

tance towards amikacin and ciprofloxacin (Ebrahimipour et al. 2018).

It is perceived that integrons are also extant in poultry. In this regard, there are several reports depicting the presence and indicting the risk of these antibiotic-resistant genes being contagious. This is evident from the study by Racewicz et al. (2022) on the assessment of resistance determinants of *E. coli* in poultry litter, faeces, and neck skin. It revealed the prevalence of class 1 integron depicting 100% resistance towards ampicillin and doxycycline in poultry litter, 100% resistance towards ampicillin and 92% resistance to ciprofloxacin in faeces and 100% resistance to ampicillin and 94.7% resistance to sulfamethoxazole in neck skin. A similar observation was made by Witaningrum et al. (2021) on *C. freundii* isolated from broiler chicken who found the presence of class 1 integrons in 22.72% of MDR *C. freundii* isolates (Witaningrum et al. 2021). Not surprisingly, *E. coli* strains containing integrons carrying different ARGs were widely studied as the possible reservoir for sulphonamide resistance genes and their dissemination will have potential risks to human health (Soufi et al. 2011).

Numerous studies have demonstrated the frequency of class 1 integron in *Enterobacteriaceae* and other pathogenic bacteria in ready-to-eat food products, water, and poultry at an alarming rate. All these reports strongly suggest that a large portion of drug resistance determinants are mainly associated with integrons with prominent ARGs. Though class 1 integrons have been widely described from clinical contexts for integron-mediated resistance, the reports of Gram-negative microbes harbouring class 2 integrons have also been established, documented, and their function in the dispersion and spread of antibiotic resistance has been validated and recognised. Similar to class 1 integrons, class 2 integrons were also detected in a diverse range of microorganisms from clinical to environmental sources and from domesticated to wild animals (Stokes and Gillings 2011, Xia et al. 2013). For instance, *Providencia vermicola* and *Proteus mirabilis* strain isolated from Salmon river, USA that harboured class 2 integrons carrying *dfrA1-sat-aadA1* gene cassettes, with the latter harbouring additional *tetA* gene conferring resistance to tetracycline (Xu et al. 2011). A metagenomic study detected potentially functional class 2 integron integrase genes in Firmicutes and Bacteroides isolated from agricultural habitats (Rodríguez Minguela et al. 2009). However, the integrase genes of class 2 integrons isolated from strains of clinical origin were found to be inactivated due to an internal stop codon (Martín et al. 2008). This further restricts the ability of these integrons to integrate gene cassettes from external sources (Gillings 2014). It was also documented that 15.3% of *P. aeruginosa* isolates from Iran carried class 2 integrons, of which 14.58% were MDR, while 0.7% were sensitive (Zarei et al. 2018). Further, *E. coli* strains isolated from broiler chickens in South Iran harboured class 2 integrons at a frequency of 2.60%, 25.50%, and 30.40% in their three rearing periods, respectively (Kalantari et al. 2021). The class 2 integron was also identified in 3 strains (20.0%) of *Shigella* species isolated from water sources with the classic gene cassette array *dfrA1-sat2-aadA1* (Shahin et al. 2019). Among Gram-positive bacteria, Methicillin-Resistant *Staphylococcus aureus* (MRSA) and Methicillin-Sensitive *Staphylococcus aureus* (MSSA) harboured class 2 integrons with a frequency of occurrence of 17.4%. The study also correlated ($P < 0.05$) the frequency of class 1 and 2 integrons and antibiotic resistance

to aminoglycosides more frequently than other antimicrobial agents (Afsharian et al. 2019).

The ubiquity of class 3 integrons is much lower than the other two classes, and hence very few studies determined their pervasiveness. The presence of class 3 integrons in *S. marcescens* was predominant in a study conducted in Poland. It consisted of 3 gene cassettes (*dfrB3*, *bla*_{OXA/aac(6)-Ib-cr}, and *bla*_{GES-7}) and this integron was carried on a novel plasmid that was named pPCMI3. This study also reported a gene *oxa/aac(6)-Ib-cr* that codes for a brand-new fusion protein made up of the enzymes OXA-lactamase and *aac(6)-Ib-cr* (Celejewski Marciniak et al. 2021). In another study in Egypt, class 3 integron was present in 13% of total integron-positive isolates (Rizk and El Mahdy et al. 2017). In accordance with a study in Nigeria, class 3 integron genes were detected in the plasmid DNA of three (27.3%) of the MDR *P. aeruginosa* strains (Eremwanarue et al. 2020). Unlike the rest of the classes, class 4 integrons are infrequent and hence very minimal studies provide evidence for these integrons. For instance, about 216 unknown genes were identified within class 4 integron in an array of 179 *V. cholerae* cassettes occupying 3% of its entire genome. It was found to carry chloramphenicol and fosfomycin resistance genes and other genes of unknown function. Apart from *Vibrionaceae*, class 4 integrons were also characterized in pseudomonads. It was identified in *Pseudomonas alcaligenes*, with the size of 18 kb consisting of 33 repeats. It also encoded for several ORFs that coded for protein of unknown function (Fluit and Schmitz 2004). However, there is a need of exploration for class 4 and 5 integrons in different *Enterobacteriaceae*. Integrons are natural cloning tools that serve as expression vectors for captured genes and agents of gene capture. Consequently, investigating their structure and diversity will help us better comprehend the regions where they can be modified. With the help of GETs currently being used, we can find a way to edit integrons and prevent them from being harmful drug resistance disseminators.

Genome Editing Technologies

Genome editing technologies allowed scientists to manipulate specific genes within numerous DNA bases in both prokaryotic and eukaryotic cells (NHGRI 2022). With the ability to insert or delete, gene manipulation provides a clear insight into the role of a particular gene, a causative gene for the disease, or even a possible prevention for a condition with a genetic cause. However, the latter is the most beneficial, as silencing genes to prevent gene expression can be a breakthrough in preventing dreadful diseases. Technologies involving Transcription Activator like Effector Nucleases (TALENs), Zinc Finger Nucleases (ZFNs), and CRISPR- Cas are the prime tools for editing an organism's genome (Gaj et al. 2013, Joung and Sander 2013). Even though TALENs and ZFNs are restriction enzymes that cleave the DNA at a specific site, they differ in the several other properties. Table 2 depicts the diversity amongst these tools, thus summarizing their pros and cons.

CRISPR-cas

Clustered regularly interspaced short palindromic repeats/CRISPR-associated protein (CRISPR-Cas) is the adaptive immune system in most archaea and bacteria. They contain a group of intercistronic, interspersed non-protein

coding that are widely distributed throughout their genome (Kundar and Gokarn 2022). The ground-breaking invention by Jennifer Doudna and Emmanuelle Charpentier has made genome editing much simpler, faster, and more efficient (Westermann et al. 2020). About 50% of sequenced bacterial genomes and almost 90% of sequenced archaeal genomes contain CRISPR. Cas, also known as “CRISPR-associated protein” is an enzyme that recognises and cleaves particular DNA strands that are complementary to CRISPR sequences (Shukla et al. 2021). CRISPR sequences together with Cas enzymes, form the basis of CRISPR-Cas technology that can be used to edit genes (Zhang et al. 2014).

The interesting study conducted by Li et al. (2020), conveyed the significance of CRISPR for combating MDR; the study reported the reduction of class 1 integrons in MDR *E. coli* by CRISPR system. These interdisciplinary studies are broadening the applications of the CRISPR system that enables the editing of MGE's like integrons resulting in the prevention of MDR dissemination amongst Gram-negative and Gram-positive bacteria. The CRISPR- Cas system is classified into two major classes: Class 1 and Class 2. Class 1 is made up of multi- subunit effector complex and is further fractionated into Type I, Type III, and Type IV while Class 2 is classified into Type II, Type V, and Type VI with single—protein effector subunit (Koonin et al. 2017) (Table 2). The CRISPR/Cas system enlists three key processes for editing target DNA: interference, expression, and adaptability. Even though the process is similar, each CRISPR- Cas system differs by the presence and absence of certain molecules and nucleases. The process begins with the guide RNA (sgRNA) formation and activation. In Type I and II system, sgRNA consists of CRISPR RNA (crRNA) that binds to trans-activating crRNA (tracrRNA) to form an active complex, while the remaining classes contains single sgRNA with crRNA. This activated complex leads to the activation of the respective Cas enzymes. sgRNA, along with Cas 9 and Cas 12, now identifies the protospacer-adjacent motif (PAM) sequence in Type II and Type V systems respectively. Cas 13 nuclease of Type VI instead identifies protospacer flanking site (PFS) whereas, Type III system doesnot require any PM site as the cleavage is independent of the sgRNA complementarity. After recognition, desired cut is made in the target DNA by the nucleases, thus leading ds/ss-DNA/RNA breaks. The genomic DNA cleaved can then be repaired using Non—Homologous End Joining (NHEJ) repair system or by Homology—Directed Repair (HDR) (Barangou and Doudna 2016, Redman et al. 2016, Jiang and Doudna 2017, Nierzwicki et al. 2020). HDR repair mechanism repairs double-stranded break (DSB) with low efficiency using a donor DNA template for gene modification, whereas the NHEJ repair mechanism repairs DSB, which results in genomic insertion or deletion with high efficiency (Barman et al. 2020).

CRISPR mediated genome editing in *Enterobacteriaceae*

Although the CRISPR mediated genome editing in *Enterobacteriaceae* was explored in the year 2015 by Jiang and his colleagues (Jiang et al. 2015), the investigations in this area have been increasing tremendously. With years passing by, the advancement in CRISPR- Cas techniques has also gradually progressed. While Jiang et al. achieved 100% efficient editing of three gene targets through CRISPR- Cas insertion and

Table 2. Comparative analysis of the different Genome editing Tools used for genome editing

GETs → Function ↓	Class 1 CRISPR/Cas system				Class 2 CRISPR/Cas system		
	ZFN	TALEN	Type I	Type III	Type IV	Type II	Type V
Target Type of target DNA cleavage	DNA, RNA Double stranded DNA break	DNA, RNA Double stranded DNA break	DNA DNA Nick	DNA, RNA Transcriptionally active DNA break and single stranded RNA break	Unknown Unknown	DNA Double stranded DNA break with blunt end	DNA Double stranded DNA break with staggered end
Endonuclease used for degradation End result	FokI	FokI	Cas 3 nuclease	Cas 10 nuclease	DinG	Cas 9 nuclease	Cas 12 nuclease
Target Efficiency Complex pattern	Insertions/deletions causing Irreversible knockout	Frame shift DNA mutation Causing Irreversible knockout	Target DNA degradation causing Irreversible gene knockout	Target DNA and RNA degradation	Unknown	Target DNA degradation leading to frameshift mutation causing permanent gene knockout/knock- in	Target DNA degradation leading to frameshift mutation causing permanent gene knockout/knock- in
Recognizes	1–50% Homodimeric and heterodimeric cleavage complex Nucleotide triplicates	1–50% Heterodimeric cleavage complex	> 70% Multiprotein cascade complex	> 70% <i>cmr</i> bound effector complex	Unknown No specific effector complexes	> 70% Binary interference complex	> 70% Binary interference complex
Guide sequence	DNA Binding domains	DNA Binding domains	Few nucleotides adjacent to PAM located at the upstream of the protospacer	Discontinuous 5 nucleotide segments	Needs more experimental study	20 base pair nucleotides adjacent to PAM sequence	Seed sequence of 3–5 nucleotides at proximal end of PAM sequence
Few Applications	Human genome editing in somatic and pluripotent stem cells	To produce chimeric antigen receptor T cells for cancer immunotherapy; plant genome editing; Genome editing of mouse, Gene knockouts in rat, mice, zebrafish, <i>Caenorhabditis elegans</i> ; Cancer cells translocation in human.	Genome editing, single and multi-gene deletions, tracrRNA	Gene silencing, Genome editing of thermophiles	plasmid propagation mechanisms, enhancement of recombination events with other nucleic acids, gene silencing	Gene insertions, deletions, replacement in bacterial genome; multiplex genome editing, modify gene expression, cancer immunotherapy,	Guide RNA (gRNA) consisting of crRNA and tracrRNA Treatment of genetic diseases, gene repression, multiplex gene regulation, bacterial genome editing, detect SARS-CoV 2 in patients sample
							development of Specific High-sensitivity Enzymatic Reporter UnLOCKing (SHERLOCK) for detection of RNA molecules, nucleotide detection in virus, RNA interference

deletion in *E. coli*, Sharma et al. (2022) generated mutagenesis in *Shigella flexneri* by generating fusion of Cas 9 nickase and cytidine deaminase that induced C to T transition resulting in the termination of the translation. They could also achieve 100% efficacy in generating loss-of-function mutants in both *S. flexneri* and *E. coli*. A modified pCas/pTargetF system was developed by Li et al. (2021) by synthesising the pEcCas/pEcgRNA system that required the replacement of promoter, original N20 sequence and change in the replicon of pCas. This updated pEcCas/pEcgRNA system was also extended to other strains of *E. coli* such as DH5 α , Nissle1917, MG1655, CGMCC3705, W strain ATCC9637 and to *Tatumella citrea* DSM13699 without any particular alteration. Eventhough the CRISPR/Cas system is the major breakthrough in the field of genome editing, effective recombination and efficient transformation is a cause of concern. However, Collias et al. (2023) demonstrated that attenuating DNA targeting activities induced *recA* mediated repair that resulted in chromosomal cleavage successfully. All these reports successfully demonstrate the effectiveness of genome editing by CRISPR- Cas9 in *Enterobacteriaceae* thereby exploring the multitude application of CRISPR- Cas9.

CRISPR as a tool to combat AMR

There have been several approaches made to solve the problem of AMR in human pathogens including the use of antisense RNA technology to silence antibiotic resistance determinants (Ashwath et al. 2022). However, CRISPR- Cas9 technique was efficiently used to generate QRDR mutations (*gyrA*83 and 87) in fluoroquinolone-resistant *E. coli* strains that reversed the resistance nature of the isolates (Qiu et al. 2018). This study further verified the role of *gyrA* mutations in quinolone resistance in *E. coli*. One of the limitations in CRISPR-mediated killing by plasmids is the decreased frequency of conjugation, which can be improved by introducing phagemid- or bacteriophage-mediated delivery systems that are much more efficient (Lossouarn et al. 2019).

There have been several studies utilising sequence-specific antimicrobials through CRISPR- Cas nuclease that employed either transformable plasmids or bacteriophages for the delivery of target molecules into drug-resistant bacteria (Bikard et al. 2014, Goma et al. 2014, Park et al. 2017). However, despite high specificity, these studies have their limitations; for instance, phage-mediated CRISPR- Cas antimicrobial dissemination showed incomplete dissemination due to selective pressure on bacteria by phages during infection and mutations of receptor regions (Duerkop et al. 2016, Chatterjee et al. 2019, Lossouarn et al. 2019). Therefore, Rodrigues et al. (2019) developed a strategy to deliver CRISPR based antimicrobial to remove erythromycin-resistant intestinal *Enterococcus faecalis* using pheromone-responsive plasmids (PRPs). The study exhibited a significant reduction in antibiotic resistance among enterococcal bacteria both *in vitro* and *in vivo*. Though the frequency of conjugation was low, the specificity of CRISPR- Cas targeting system *in vivo* was absolute. Nevertheless, these techniques could target one antibiotic-resistance gene at a time; hence the possibility of other mechanisms of resistance exhibited by the bacteria for the same antibiotic cannot be neglected. Since integrons are major mobile genetics elements that can carry multiple resistance gene cassettes expressed by a single promoter, they can be a potential target for CRISPR-Cas-based sequence-specific antimicrobial

activity. Also, due to the high specificity, high accuracy, cost effectiveness, less laborious nature of CRISPR- Cas system, it could be the most promising genome editing tool for the same. There have been tremendous advances in gene editing using CRISPR- Cas9 to modify the genome of a microorganism. Zhao et al. (2016) developed a fast and easy method for genome editing with one plasmid construction and a single transformation step in *E. coli*. This study highlights the lesser time consumed by CRISPR- Cas9 in comparison to other methods. Wang et al. (2016) performed genome editing by creating deletion, integration, and single nucleotide modification, thus producing a clean mutant in *Clostridium beijerinckii*. The unique feature of their study was that they modified a single nucleotide without the necessity of PAM sequences. Not just in bacteria, but this unique genome editing tool has been proven efficient in plant systems too. Target gene knockout has been successfully performed in rice, arabidopsis, sorghum, and tobacco (Kuhn and Binder 2002, Feng et al. 2013, Mao et al. 2013). A precise CRISPR- Cas9 system technique for targeted mutagenesis in wheat and rice was also described by Shan et al. (2014). A recent study by Ma et al. (2020) showed the use of a negative-strand RNA virus-based vector to deliver CRISPR- Cas9 cassette. This study achieved the goal of DNA-free cassette delivery along with chromosome deletions and single, multiplex mutagenesis at high frequency in a tobacco host. All these studies reflect the success rate of use of CRISPR- Cas system for genome editing, which could be a prominent evidence to adopt CRISPR- Cas9.

A sequence-specific gRNA against the promoter region of the integron can be designed and cloned into pCRISPR vector. This vector, along with pCas9 will constitutively express Cas9 and can be used as an editing tool. Further a repair template with left and right homology arms that are complementary to the target DNA fragment can be used to replace the promoter region with the help of lambda red gene construct that initiates recombination (Fig. 2). With this mechanism, the entire structure of an integron can be made inactive, thereby making it possible to use antibiotics that were earlier thought to be not useful for treating infections caused by MDR pathogens.

The technique addressed in this review is first of its kind that focusses on targeting the promoter region of the integron to silence the ARGs within it. The proposed technique can be initially scrutinized for the successful editing of the promoter region in one of the *Enterobacteriaceae* isolate. On the successful editing, *in vivo* studies can be performed so as to check for the effectiveness of the editing in the antibiotic stress environment by evaluating the expression of the gene cassettes encoding ARGs. On the validation of this technique using *in vivo* studies, the vectors can be administered using a medium such as nanoparticles, protein conjugates carrying CRISPR/Cas vector, which on entry in to the gut will release these vectors into the external environment. For instance, a cationic polymer based nanosized CRISPR complex harbouring cas proteins and crRNA targeting methicillin resistance gene was successfully introduced in methicillin resistant *Staphylococcus aureus* (MRSA) (Kang et al. 2017). However, the rate of HGT of these vectors into the gut microbes requires more investigation.

This CRISPR- Cas9 system has proved to be fast and more efficient, yet it has some drawbacks. One of the drawbacks is the requirement for a PAM sequence near the target site. One of the most widely utilised Cas9s, SpCas9, is derived from the bacteria *Streptococcus pyogenes*, and consists of PAM

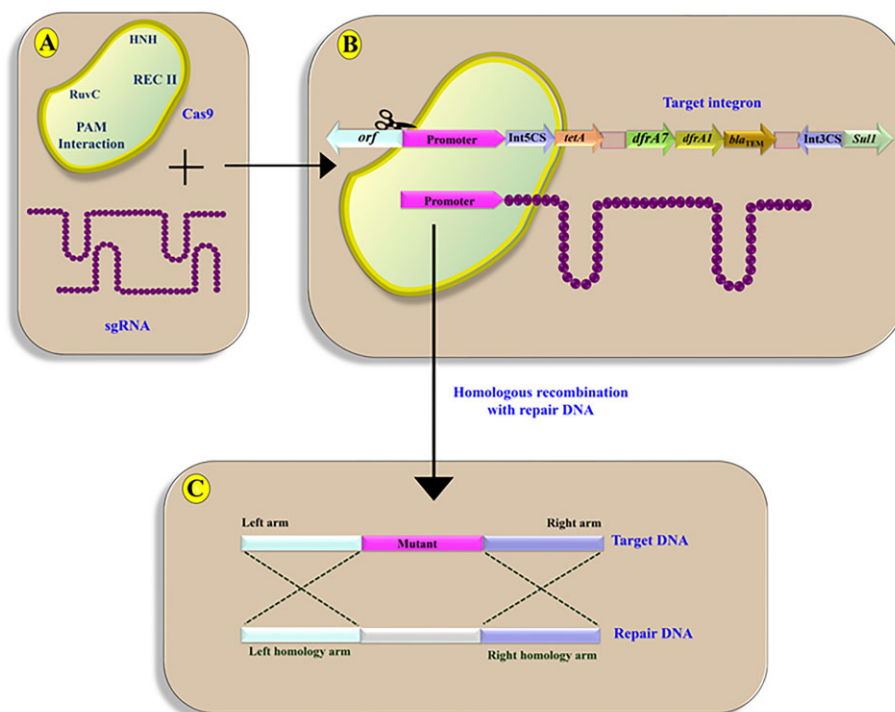


Figure 2. Schematic illustration of gene promoter silencing using CRISPR- Cas9 genome editing tool: The initial step in CRISPR- Cas9 based genome editing involves identifying and locating target genome. (A) This is achieved by the interaction of Cas9 protein with guide RNA to form an activated complex. The PAM interaction subunit in Cas9 is responsible to initiate the binding of the activated Cas9 to the target promoter. The HNH and RuvC nucleases are responsible to cleave the target promoter. (B) The activated Cas9 complex binds to promoter region by locating the PAM sequence adjacent to the target site. Once identified, the cut is induced at the target promoter. This is then followed by (C) Homologous recombination of the target promoter DNA with the repair DNA. The left and right homology arms consist of sequence complementary to the promoter DNA. The final mutated promoter obtained is now unfunctional and thus expression of the gene cassettes of this promoter will be inhibited.

recognition site 5'NGG3'. However, this Cas9 is large and challenging to package into AAV (Adeno Associated viral) vectors, thus making the requirement of PAM sequence complex (Lino et al. 2018, Singh et al. 2021). To overcome this, SaCas9, Cas9 from *S. aureus* could be used, but has a longer PAM sequence: 5'NNGRRT3'. Furthermore, DSBs induced after CRISPR- based editing triggers apoptosis (Hu et al. 2014). In addition, the challenges in the delivery of CRISPR-Cas systems into the target regions should also be considered in the future studies to eradicate AMR pathogens (Duan et al. 2021). These limitations are really of serious concern and have to be studied more precisely. Also, there have to be more number of studies on combating AMR using CRISPR, to make it target-specific, without any additional and nonspecific changes.

Conclusion

Middle and low income countries faces a considerable challenge in preventing and controlling the spread of drug resistance in *Enterobacteriaceae*. The widespread of MDR to PDR has pushed toward alternative treatment approaches. Though, there have been several attempts to develop newer technologies to curb the infection caused by drug-resistant human pathogens, the associated limitations have hindered the development of these technologies. Understanding the potential of integron by silencing the genes harboured on mobile genetic elements can be achieved by CRISPR. CRISPR- Cas can be a game changer among all the existing genome editing techniques in combating drug resistance. The CRISPR-

Cas technique has been widely used as a gene editing tool in eukaryotes. However, its importance in editing the prokaryotic genome is poorly understood. In this review, we have described the role of integrons in drug resistance in *Enterobacteriaceae* and various features of CRISPR- Cas technology which has been widely used for genome editing. Almost all the classes of antibiotics have lost their susceptibility toward Gram-negative organisms in one or the other ways. Therefore, the CRISPR- Cas9 system can be the best way to revert the resistant isolates to sensitive ones. Since, integrons are important genetic structures in prokaryotes that mainly contribute to the drug-resistance characteristics of human pathogens, they can be a potential target to utilise CRISPR- Cas9 technology in the prevention and dissemination of drug resistance determinants.

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Conflicts of interest

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Author contributions

Varsha Prakash Shetty (Conceptualization, Data curation, Formal analysis, Investigation, Validation, Writing – original draft, Writing – review & editing), Sadanand Dangari Akshay (Formal analysis, Investigation, Methodology, Writing – review & editing), Praveen Rai (Data curation, Investigation, Validation, Writing – review & editing), and Vijaya Kumar Deekshit (Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Resources, Supervision, Visualization, Writing – original draft, Writing – review & editing)

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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